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Engineering Mathematics

4Week [25 September - 01 October]

Quiz 4

Started on	2023-10-02 08:00
State	Finished
Completed on	2023-10-02 09:00
Time taken	59 mins 11 secs
Grade	7.00 out of 10.00 (70%)

Question 1

Correct

Mark 1.00 out of 1.00

Choose the second order homogeneous linear differential equation with constant coefficients from the list below.

Select one:

- ☒ $y'' + 2y' - y = 0$ ✓
- ☐ $y'' + 2y' - y = \sin t$
- ☐ $y'' + 2(y')^2 - y = 0$
- ☐ $y'' + 2ty' - y = 0$

Your answer is correct.

The correct answer is: $y'' + 2y' - y = 0$

Question 2

Correct

Mark 1.00 out of 1.00

A 2kg mass is attached to a spring with stiffness $k=50$ N/m. The mass is displaced $1/4$ m to the left of the equilibrium point and given a velocity of 1 m/sec to the left. Neglecting damping, find the equation of motion of the mass.

Select one:

- ☒ $2y'' + 50y = 0, \quad y(0) = -1/4, \quad y'(0) = -1$ ✓
- ☐ $2y'' - 50y = 0, \quad y(0) = 1/4, \quad y'(0) = -1$
- ☐ $50y'' + 2y = 0, \quad y(0) = -1/4, \quad y'(0) = -1$
- ☐ $50y'' - 2y = 0, \quad y(0) = 1/4, \quad y'(0) = -1$

Your answer is correct.

The correct answer is:

$2y'' + 50y = 0, \quad y(0) = -1/4, \quad y'(0) = -1$

Question 3

Correct

Mark 1.00 out of 1.00

Choose a function which is a solution of the equation

$$my'' + ky = 0$$

Select one:

- ☒ $y = \cos wt$, where $w = \sqrt{\frac{k}{m}}$ ✓
- ☐ $y = e^{wt}$, where $w = \sqrt{\frac{k}{m}}$
- ☐ $y = e^{-wt}$, where $w = \sqrt{\frac{k}{m}}$
- ☐ $y = te^{-wt}$, where $w = \sqrt{\frac{k}{m}}$

Your answer is correct.

The correct answer is: $y = \cos wt$, where $w = \sqrt{\frac{k}{m}}$

Question 4

Correct

Mark 1.00 out of 1.00

Choose a pair of functions which are linearly independent on $(0, 1)$.

Select one:

- ☒ e^{3t} and e^{-4t} ✓
- ☐ e^{2t} and $5e^{2t}$
- ☐ $\tan^2 t - \sec^2 t$ and 3
- ☐ $\cos t \sin t$ and $\sin 2t$

Your answer is correct.

The correct answer is: e^{3t} and e^{-4t}

Question 5

Correct

Mark 1.00 out of 1.00

Find the general solution of $y'' + 2y' - 8y = 0$.

Select one:

- ☒ $y = c_1 e^{2t} + c_2 e^{-4t}$ ✓
- ☐ $y = c_1 e^{4t} + c_2 e^{-2t}$
- ☐ $y = c_1 e^{2t} + c_2 e^{-2t}$
- ☐ $y = c_1 e^{4t} + c_2 e^{-4t}$

Your answer is correct.

The correct answer is: $y = c_1 e^{2t} + c_2 e^{-4t}$

Question 6

Correct

Mark 1.00 out of 1.00

Find the general solution of $36y'' - 12y' + 37y = 0$.

Select one:

- ☒ $y = c_1 e^{t/6} \cos t + c_2 e^{t/6} \sin t$ ✓
- ☐ $y = c_1 e^t + c_2 e^{t/6} \sin t$
- ☐ $y = c_1 e^{-t/6} \cos t + c_2 e^{-t/6} \sin t$
- ☐ $y = c_1 e^t \cos(t/6) + c_2 e^t \sin(t/6)$

Your answer is correct.

The correct answer is: $y = c_1 e^{t/6} \cos t + c_2 e^{t/6} \sin t$

Question 7

Correct

Mark 1.00 out of 1.00

Find the general solution of $y'' - 12y' + 36y = 0$.

Select one:

- ☒ $y = c_1 e^{6t} + c_2 t e^{6t}$ ✓
- ☐ $y = c_1 e^{-6t} + c_2 t e^{-6t}$
- ☐ $y = c_1 e^{6t} + c_2 e^{7t}$
- ☐ $y = c_1 e^{6t} + c_2 e^{6t}$

Your answer is correct.

The correct answer is: $y = c_1 e^{6t} + c_2 t e^{6t}$

Question 8

Incorrect

Mark 0.00 out of 1.00

A 1/8 kg mass is attached to a spring with stiffness $k=16$ N/m. The mass is displaced 1/2 m to the right of the equilibrium point and given an outward velocity (to the right) of 22 m/sec. Ignoring the damping coefficient, determine the period of motion.

Select one:

- ☐ $\frac{\sqrt{2}\pi}{8}$
- ☒ $\frac{\sqrt{2}\pi}{4}$ ✗
- ☐ $\frac{\sqrt{2}\pi}{2}$
- ☐ $\frac{\sqrt{2}\pi}{6}$

Your answer is incorrect.

The correct answer is: $\frac{\sqrt{2}\pi}{8}$

Question 9

Incorrect

Mark 0.00 out of 2.00

A $1/8$ kg mass is attached to a spring with stiffness $k=16$ N/m. The mass is displaced $1/2$ m to the right of the equilibrium point and given an outward velocity (to the right) of $\sqrt{2}$ m/sec. Neglecting any damping or external forces that may be present, determine the equation of motion of the mass.

Select one:

- ☐ $y(t) = \frac{\cos(8\sqrt{2}t)}{2} + \frac{\sin(8\sqrt{2}t)}{8}$
- ☐ $y(t) = \frac{\cos(8\sqrt{2}t)}{8} + \frac{\sin(8\sqrt{2}t)}{2}$
- ☒ $y(t) = \frac{\cos(4\sqrt{2}t)}{2} + \frac{\sin(4\sqrt{2}t)}{8}$ ✖
- ☐ $y(t) = \frac{\cos(4\sqrt{2}t)}{8} + \frac{\sin(4\sqrt{2}t)}{2}$

Your answer is incorrect.

The correct answer is: $y(t) = \frac{\cos(8\sqrt{2}t)}{2} + \frac{\sin(8\sqrt{2}t)}{8}$